SURJECTIVITY ONTO EACH ONE-GENERATOR COMPLETION IMPLIES SURJECTIVITY ONTO THE IDEAL-ADIC COMPLETION

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file algebra.tex; no statement or proof step was rewritten, reordered or removed.

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Allowed standard prerequisites (editorial).

- Comparison of the I-adic tower with the tower cut out by the powers of the generators. Used at the first line of the counted proof (line 22980). Justifies replacing the completion along I by the completion along the r generators, up to the same inverse limit.
- The symbolic writing of an element of the limit. Used in the counted proof at lines 22982-22986. Represents a compatible family as a sum of f_i^n x_{n,i}, which can then be matched componentwise.

Lemma 0.1. *Let A be a ring. Let $I = (f_1, \ldots, f_r)$ be a finitely generated ideal. If $M \to \lim M/f_i^n M$ is surjective for each i, then $M \to \lim M/I^n M$ is surjective.*

Proof. Note that $\lim M/I^n M = \lim M/(f_1^n, \ldots, f_r^n)M$ as $I^n \supset (f_1^n, \ldots, f_r^n) \supset I^{rn}$. An element ξ of $\lim M/(f_1^n, \ldots, f_r^n)M$ can be symbolically written as

$$\xi = \sum\nolimits_{n \geq 0} \sum\nolimits_i f_i^n x_{n,i}$$

with $x_{n,i} \in M$. If $M \to \lim M/f_i^n M$ is surjective, then there is an $x_i \in M$ mapping to $\sum x_{n,i} f_i^n$ in $\lim M/f_i^n M$. Then $x = \sum x_i$ maps to ξ in $\lim M/I^n M$. □

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